COMP S492F/3920SEF Machine Learning

Real-world data

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Department of Electronic Engineering and Computer Science (EECS)
School of Science and Technology (S&T)
Hong Kong Metropolitan University

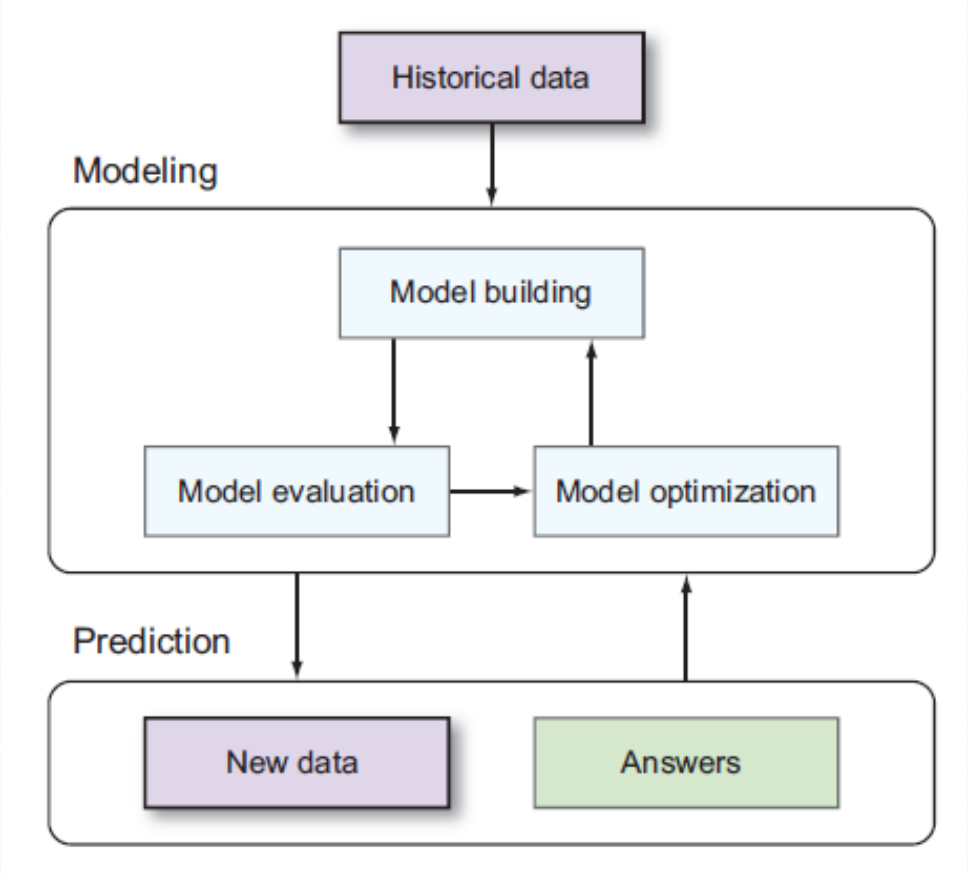
Contents

- Getting started: data collection
- Data preprocessing
- Optimization

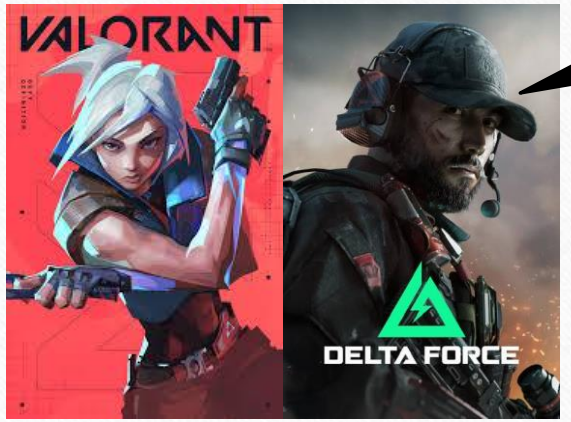
Getting started:

data collection

basic ML workflow



PUBG Finish Placement Prediction



The teacher is outdated

Titanic Survival Prediction

Take PUBG as an example

Take PUBG as an example

What are the commonalities in these questions?

- Will this player make it to the top 10 in this match?
- Can this squad (team) survive the first safe zone collapse?
- Is this player likely to achieve 5 or more kills?
- Will the player who lands in Camp Bravo get eliminated within the first 5 minutes?
- What will be final placement for this player?

➢ **Making assessments on one or several instances of interest**
(e.g., a specific player or squad in a specific match)
➢ **Has a well-defined target of interest**
(e.g., Top 10 Finish, Survival Time, Kill Count, Placement)
➢ **Has sets of historical data in which the target is known**
(e.g., thousands of past matches with recorded player stats and final outcomes)
➢ **Comes with an implied action if the target were knowable**
(e.g., adjust matchmaking, personalize in-game challenges or rewards, provide real-time tactical suggestions)

PUBG Finish Placement Prediction

What will be the finish placement for each player?

instance/example

Id	assists	boosts	Damage Dealt	heals	killPlace	kills	Match Duration	Ride Distance	Walk Distance	Weapons Acquired	**winPlace Perc**
1	0	0	0.0	1	72	0	1294	0.00	1628.00	2	**0.4446**
2	0	0	100.0	0	56	1	1294	244.80	2530.00	5	**0.9660**
3	0	1	0.0	0	75	0	1294	0.00	581.00	1	**0.2233**
4	0	0	0.0	0	89	0	1294	0.00	0.00	0	**0.0000**

Training data for PUBG Finish Placement Prediction

- Which input features should I include?
- How much training data do I need?

Which input features should I include?

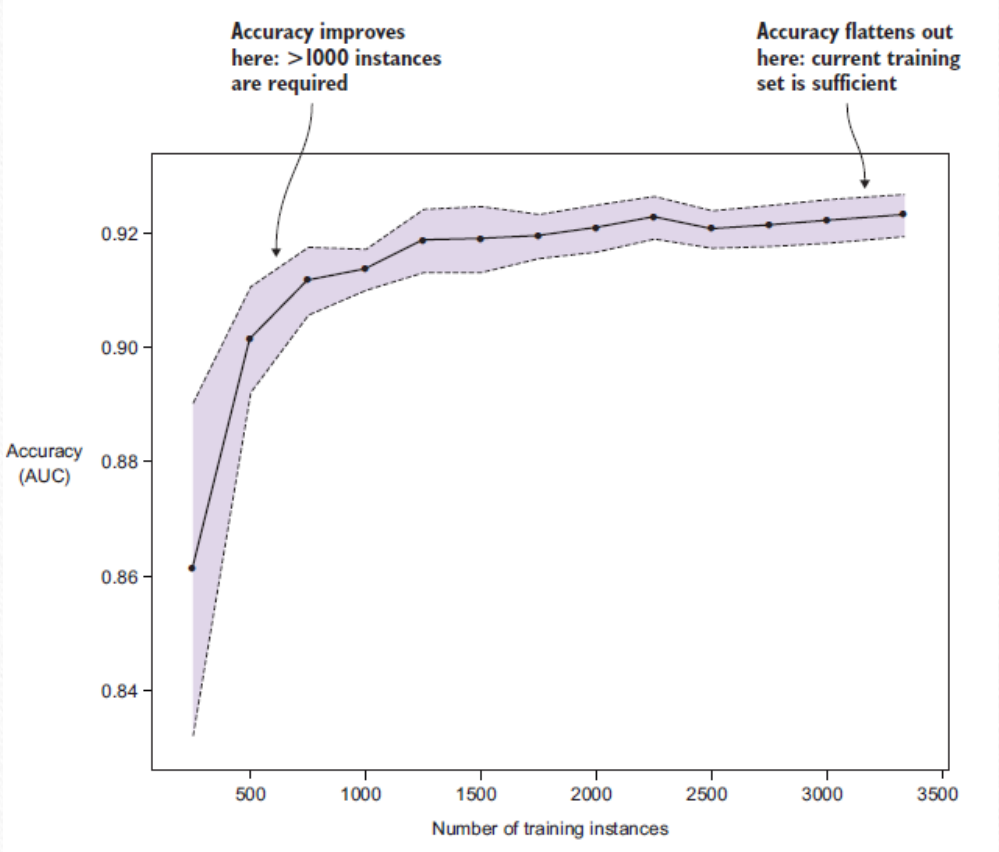
Features											Target
Id	**assists**	**boosts**	**Damage Dealt**	**heals**	**killPlace**	**kills**	**Match Duration**	**Ride Distance**	**Walk Distance**	**Weapons Acquired**	**winPlace Perc**
1	0	0	0.0	0	72	0	1294	0.00	1628.00	2	**0.4446**
2	0	0	100.0	1	56	1	1294	244.80	2530.00	5	**0.9660**
3	0	1	0.0	0	75	0	1294	0.00	581.00	1	**0.2233**
4	0	0	0.0	0	89	0	1294	0.00	0.00	0	**0.0000**

- Features that obviously have nothing to do with the target should be excluded.
- The more uninformative features are present, the less accurate the ML model will be.

How much training data do I need?

- The complexity of the problem. Relationship between the input features and target variable.
- The requirements for accuracy.
- The dimensionality of the feature space.

How much training data do I need? (Con't)



1. Choose a grid of subsample sizes to try (e.g., 250,500,750,1000,…).
2. Randomly draw the instances for each subsample size.
3. Build an ML model and assess the accuracy of that model for each subsample size. Repeat this step to get an average value of accuracy.
4. Calculate the change of average accuracy.

Data preprocessing

Take Titanic survival prediction as an example

Categorical features

simple Person dataset

Person	Name	Age	Income	Marital status
1	Jane Doe	24	81,200	Single
2	John Smith	41	121,000	Married

Categorical features

Titanic passengers

PassengerId	Survived	Pclass	Gender	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Male	22	1	0	A/5 21171	7.25		S
2	1	1	Female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Female	26	0	0	STON/02. 3101282	7.925		S
4	1	1	Female	35	1	0	113803	53.1	C123	S
5	0	3	Male	35	0	0	373450	8.05		S
6	0	3	Male		0	0	330877	8.4583		Q

Number of siblings/spouse aboard

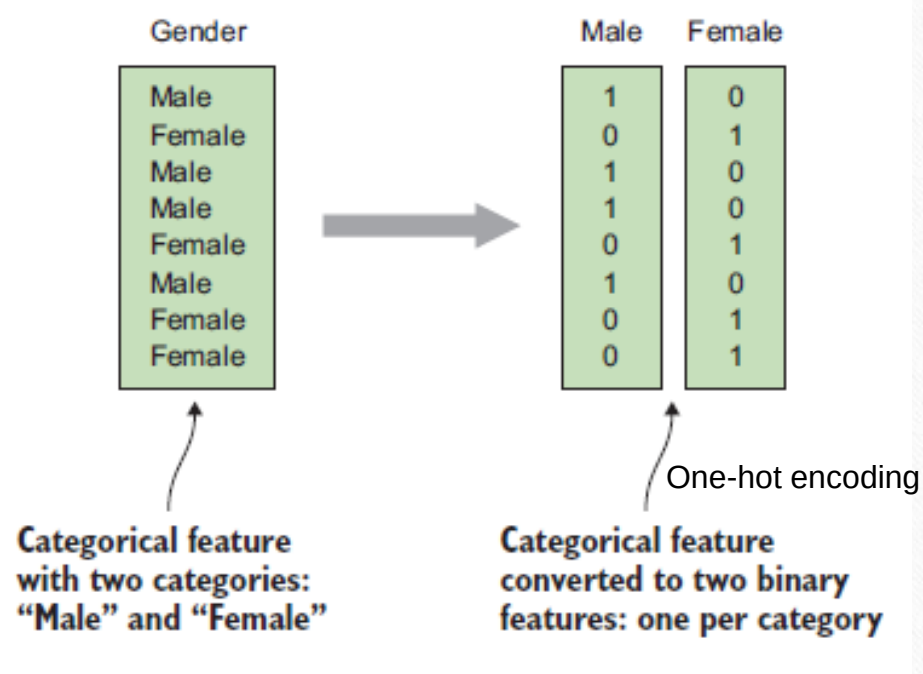
Number of parents/children aboard

Port of embarkation

- **Female survival rate**: ~74%
- **Male survival rate**: ~19%
- **1st class survival rate**: ~63%
- **3rd class survival rate**: ~24%
- **Children survival rate**: ~59%

Categorical features

Converting categorical columns to numerical columns



How we deal with gender in the USA?

Male
Female
Transgender
Genderqueer-酷儿性别
Genderfluid-流性人
Agender-无性别者
Bigender-双性别者
… …

Categorical features

How we deal with gender in the USA?

Male	Female	Transgender	Genderqueer	Genderfluid	Agender	Bigender
1	0	0	0	0	0	0
0	1	0	0	0	0	0
0	0	1	0	0	0	0
0	0	0	1	0	0	0

… …

OR

Male	0
Female	1
Transgender	2
Genderqueer	3
Genderfluid	4
Agender	5
Bigender	6

What are the pros and cons of the two methods?

Missing data

PassengerId	Survived	Pclass	Gender	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Male	22	1	0	A/5 21171	7.25		S
2	1	1	Female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Female	26	0	0	STON/02. 3101282	7.925		S
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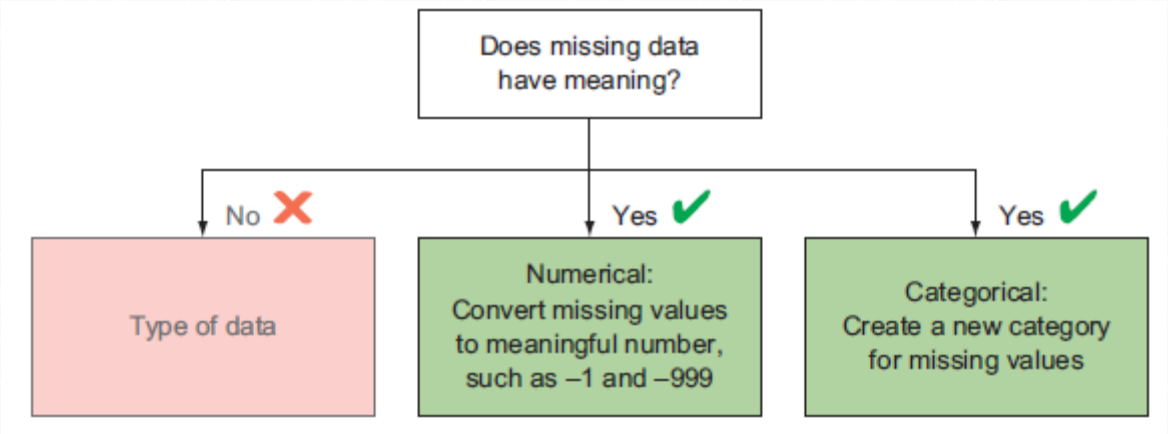
5	0	3	Male	35	0	0	373450	8.05		S
6	0	3	Male		0	0	330877	8.4583		Q

Missing values

The Titanic Passengers dataset has missing values in the Age and Cabin columns. The passenger information has been extracted from various historical sources, so in this case the missing values stem from information that could not be found in the sources.

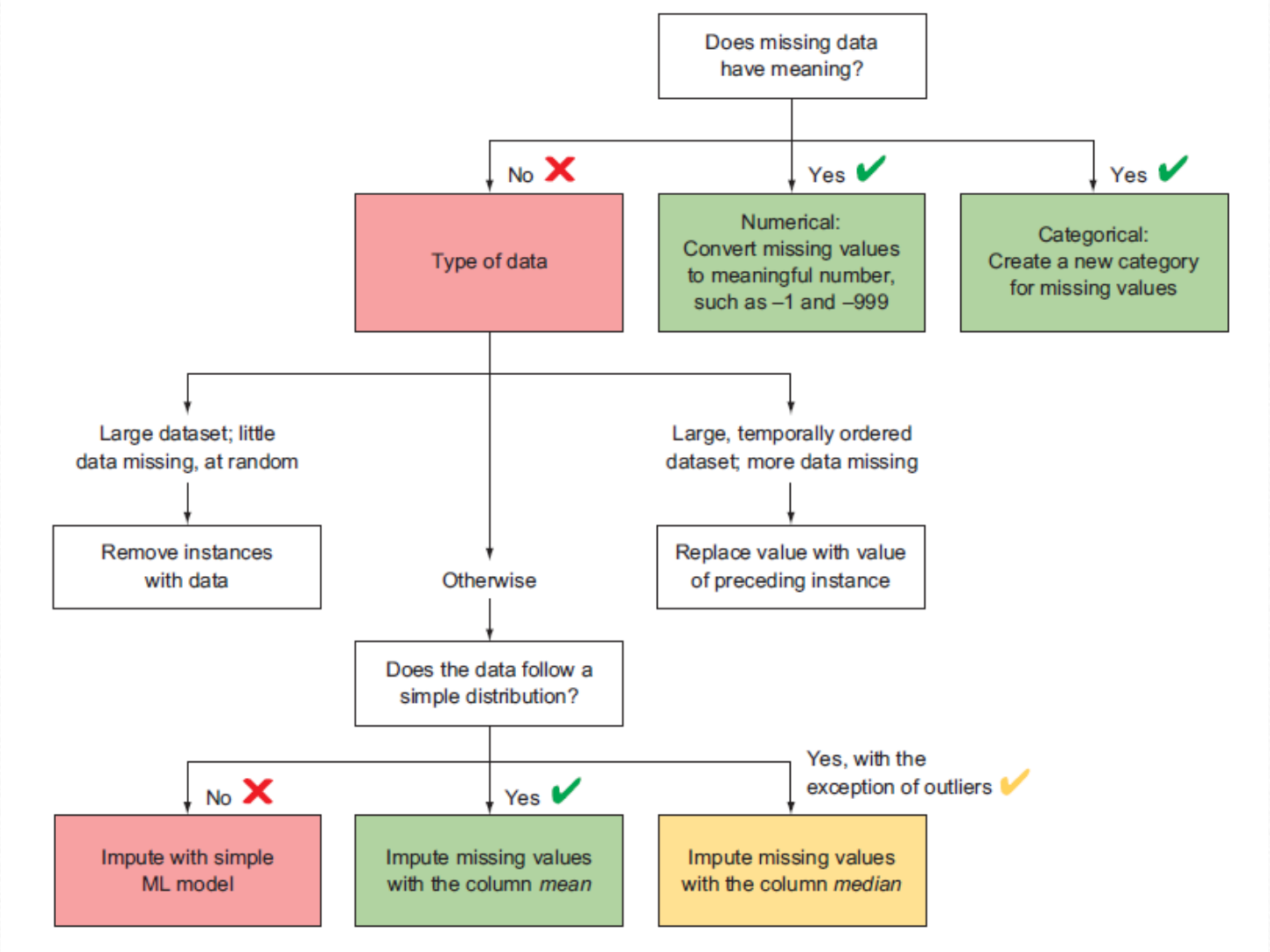
Missing data

For informative missing data:



For a data item has no informative value:

- eliminate missing values
- or replace them by guessing the true value (**imputation**)



Full decision diagram for handling missing values when preparing data for ML modeling

Data normalization

- .The value range of a feature can influence the importance of the feature compared to other features.
  - E.g., one feature has values between 0 and 10, and another has values between 0 and 1
- Often data is normalized to be in the range from 0 to 1, or from –1 to 1.

Feature normalization

```
def normalize_feature(data, f_min=-1.0, f_max=1.0):
    d_min, d_max = min(data), max(data)
    factor = (f_max - f_min) / (d_max - d_min)
    normalized = f_min + (data - d_min)*factor
    return normalized, factor
```

Optimization

- What is optimization?
- Terminology and notations
- Summary

What is optimization?

- When we solve an optimization problem, we try to find out the best solution. E.g. :
  - Problem 1, find the cheapest flight from HK to London;
  - Problem 2, find the cheapest flight from HK to London which will take off at night ("take off at night" here is a constraint of the problem, it is a condition that is required to be satisfied).
  - Problem 3, minimize the route distance from Yuen Long to Kowloon.
  - Problem 4, minimize the route distance from Yuen Long to Kowloon by taxi.

- An optimization problem may have no constraints (unconstrained optimization problem), or have at least one constraint (constrained optimization problem).

Terminology and notations

- E.g., minimize the route distance from Yuen Long to Kowloon by taxi.

	variable	objective function	constraint
Denoted by:	x	$f(x)$	$h(x)$ or/and $g(x)$

- The objective function, $f(x)$, which is the output we are trying to minimize.
- x is a variable (or decision variable) of $f(x)$. It is the input of $f(x)$ and it is the thing we can control. There may be more than one variables.
- Ideally, the route distance should be the displacement between Yuen Long and Kowloon. However, the constraint imposes a limitation (by taxi) to the problem.
- $h(x)$ and $g(x)$ are the constraints of $f(x)$. We usually denote $h(x)$ as equality constraints and $g(x)$ as inequality constraints.

Terminology and notations (Con't)

- Formulation of a constrained optimization problem:

$$\min_{x} f(x)$$

$$\begin{aligned} subject\ to\ h_i(x) &= 0, \qquad i = 1,2,\dots,m \\ g_j(x) &\leq 0, \qquad j = 1,2,\dots,p \end{aligned}$$

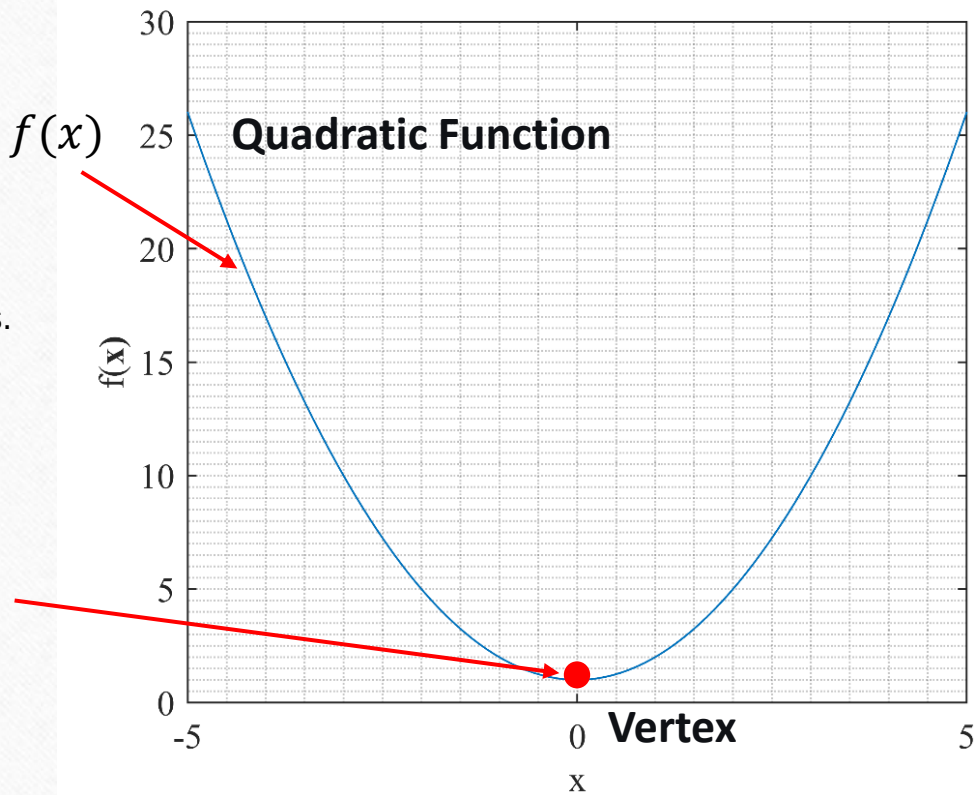
Example 1

$$\min_{x} f(x) = x^2 + 1$$
$$subject\ to\ -5 \leq x \leq 5$$

As the variable is continuous,
there are infinite feasible solutions.

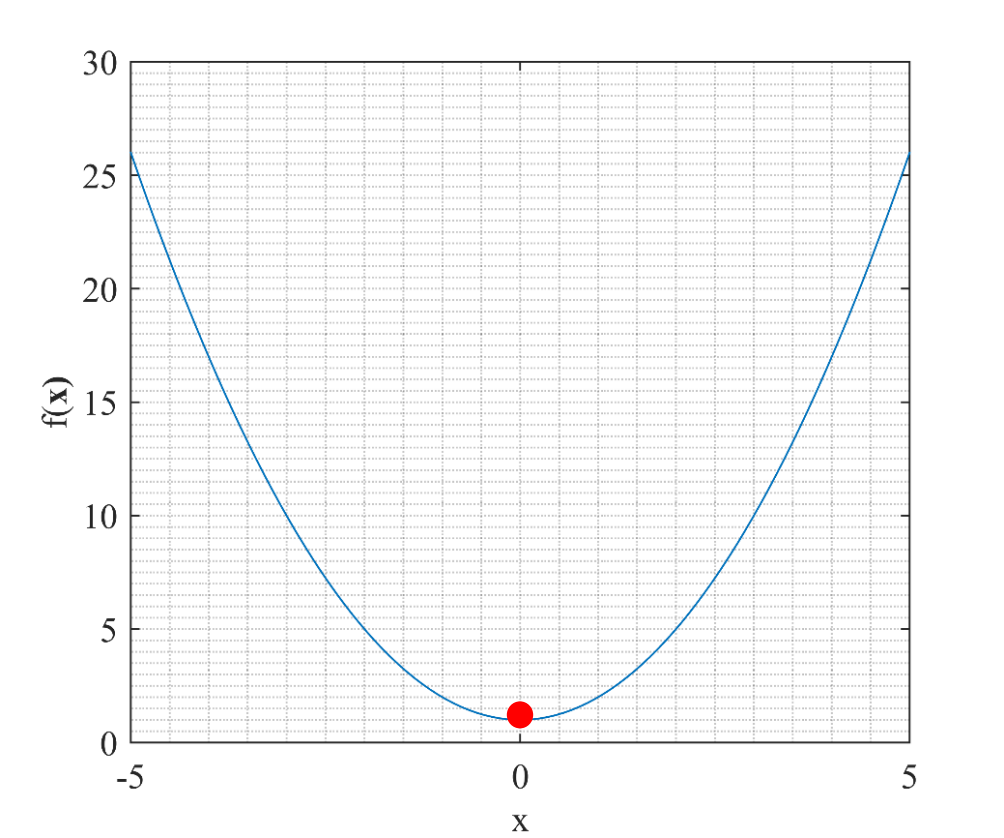
$f(x) = 0 * 0 + 1$
$= 1$

We say that $x = 0$ is the global minimum, and the minimum value of the objective function is 1 (i.e., f(x)=1).



Example 1 (Con't)

$$\min_{x} f(x) = x^2 + 1$$
$$subject\ to\ -5 \leq x \leq 5$$



- The goal of an optimization problem is to find the best solution from a set of feasible solutions.
- A feasible solution should satisfy all of the constraints in an optimization problem.

Example 2

$$\min_{x} f(x) = x_1 + x_2$$

$$subject\ to\ x_1 \geq 2$$

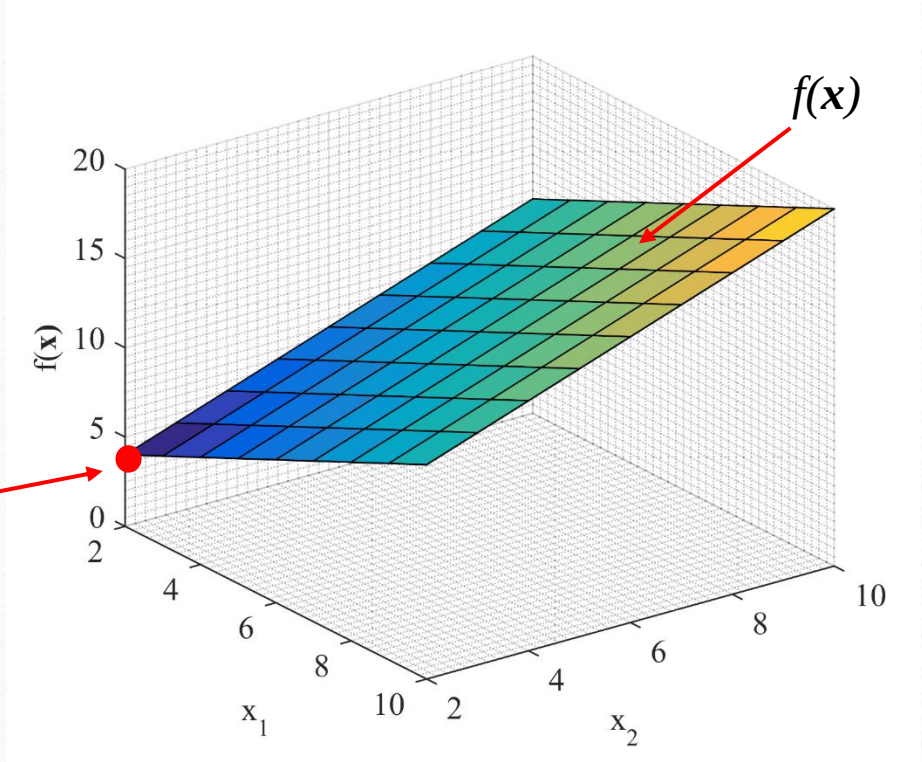
$$x_2 \geq 2$$

where $x = (x_1, x_2)$.

$f(x) = 2 + 2$

$= 4$

We say that $x = (2,2)$ is the global minimum, and the minimum value of the objective function is 4.



Example 3

$$\min_x f(x) = x^4 + x^3 - 3x^2 - 2x$$
$$subject\ to - 2 \leq x \leq 2$$

$How\ we\ find\ out\ the\ minimal\ or\ maximal\ values\ (Extrema)?$

First derivative:

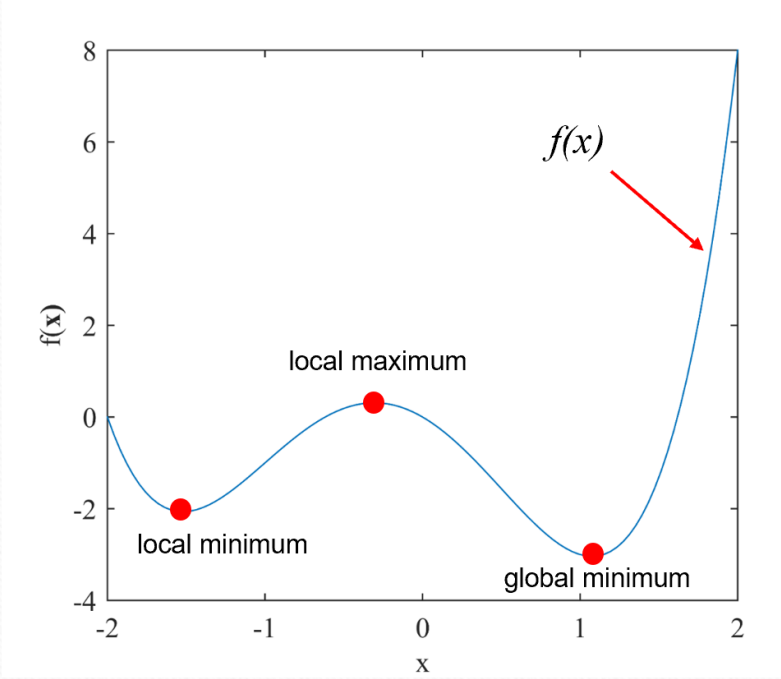
$$f'^{(x)} = 4x^3 - 3x^2 - 6x - 2 = 0$$

$How\ we\ find\ out\ which\ of\ the\ Extrema\ are\ minimal\ or\ maximal?$

Second derivative:

$$f''^{(x)} = 12x^2 - 6x - 6 > 0 \qquad minimal$$

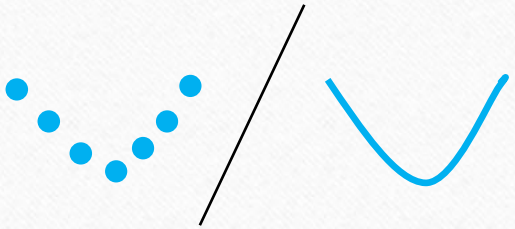
$$f''^{(x)} = 12x^2 - 6x - 6 < 0 \qquad maximal$$



The maximum or minimum over the whole function is called a global maximum or minimum.
There is only one global maximum or minimum but there can be more than one local maximum or minimum.

Summary

- Constrained optimization problems and unconstrained optimization problems
- An optimization problem can consists of one variable or many.
- Variables can be discrete (e.g., integer values) or continuous.



- A maximum or minimum is said to be global if it is the largest or smallest value of the function, respectively.

The end